Lab 2: Network Configuration and Connectivity

Objective

The purpose of this lab was to explore basic network configuration, connectivity, name resolution, and IPv6 networking tools. Various command-line utilities were used to examine network settings, verify communication between hosts, analyze hostname resolution, and observe IPv6 configuration.

Lab 2.1 – Name Resolution in Operating System

Lab A – Localhost

System Information

The lab was completed on a Windows 11 system using Command Prompt.

ipconfig /all

The ipconfig /all command was executed to display detailed TCP/IP configuration information.

Results

- Host Name: LAPTOP-Q8KU3PPL
- Node Type: Hybrid
- IP Routing Enabled: No
- WINS Proxy Enabled: No
- VirtualBox Host-Only Adapter IPv4 Address: 192.168.56.1
- Subnet Mask: 255.255.255.0
- Wireless Adapter: Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC

Analysis

The system contained multiple network adapters, including a wireless adapter and a VirtualBox Host-Only adapter. DHCP was disabled on the VirtualBox adapter because it uses a statically assigned private address. The computer was not configured as a router because IP routing was disabled.

IPv4 Loopback Test

Command:

ping 127.0.0.1

Results

- Packets Sent: 4
- Packets Received: 4
- Packets Lost: 0
- Packet Loss: 0%

Analysis

The address 127.0.0.1 is the IPv4 loopback address. Successful responses confirmed that the local IPv4 TCP/IP stack was functioning properly.

IPv6 Loopback Test

Command:

ping ::1

Results

- Packets Sent: 4
- Packets Received: 4
- Packets Lost: 0
- Packet Loss: 0%

Analysis

The address ::1 is the IPv6 loopback address. Successful responses confirmed that the IPv6 protocol stack was operational.

Are the results the same? Why?

Both commands returned successful replies because they test communication with the local computer rather than another device on the network. The difference is that 127.0.0.1 uses IPv4 while ::1 uses IPv6.

Localhost Resolution

Command:

ping localhost

Results

The hostname localhost resolved to the IPv6 loopback address ::1.

Analysis

The operating system recognizes localhost as a reserved hostname. Windows resolves localhost to the loopback interface using internal configuration and local host name resolution mechanisms.

Hosts File Examination

The hosts file was opened and examined.

Existing Entries

The file primarily contained Microsoft documentation and examples. Active entries included:

- 192.168.1.2 host.docker.internal
- 192.168.1.2 gateway.docker.internal
- 127.0.0.1 kubernetes.docker.internal

Why are instructions commented out?

Lines beginning with the # symbol are comments. Comments provide documentation and instructions but are ignored by the operating system during hostname resolution.

Modifying the Hosts File

The following entries were added:

192.168.1.1 winserv

192.168.1.10 me

These entries mapped the default gateway and local computer address to custom hostnames.

Ping Test for “me”

Command:

ping me

Results

The hostname resolved to 192.168.1.10.

- Packets Sent: 4
- Packets Received: 4
- Packet Loss: 0%

Analysis

The successful resolution demonstrated that Windows consulted the hosts file and correctly translated the hostname to the assigned IPv4 address.

Ping Test for “winserv”

Command:

ping winserv

Results

The hostname resolved to 192.168.1.1.

- Packets Sent: 4
- Packets Received: 4
- Packet Loss: 0%
- Average Response Time: 1 ms

Analysis

The custom hostname successfully resolved to the default gateway, confirming proper operation of the hosts file entries.

Question 10

Can this file be exploited without administrator privileges?

Generally, the hosts file cannot be modified without administrator privileges. However, malware running with elevated privileges could alter entries and redirect users to malicious websites by changing hostname-to-IP mappings.

DNS Cache

Command:

ipconfig /displaydns

Results

The command returned:

"The requested operation requires elevation."

Analysis

The DNS cache could not be displayed because the Command Prompt session was not running with administrative privileges.

Note

Lab B Step 3 was not completed because elevated permissions were required.

Lab B – nslookup and netstat

nslookup

Command:

nslookup

Results

Default Server: openrg.home

Address: fd22:4e39:e630:1:2ede:32ff:fee2:86ff

Question: Which option displays the current server and host?

Within nslookup, the command:

set all

displays current options, the active host, and the DNS server configuration.

Analysis

The system was configured to use an IPv6 DNS server named openrg.home. The nslookup utility is used to query DNS records and troubleshoot name-resolution issues.

---

netstat /?

Command:

netstat /?

What is the function of the command?

Netstat displays protocol statistics, routing information, active network connections, listening ports, and network-related process information.

What option identifies the process creating a network connection?

The option:

netstat -o

shows the Process ID (PID) associated with each connection.

The option:

netstat -b

attempts to display the executable responsible for creating the connection.

Netstat Connections

Lab B Step 3 was not completed; therefore active network connections were not recorded.

Lab 2.2 – IPv6 Basics

Lab A – netsh Command

IPv6 Address Information

Command:

netsh interface ipv6 show address

Results

Wi-Fi Interface (Interface 5) contained multiple IPv6 addresses:

Temporary Addresses:

- fd22:4e39:e630:1:965:548f:a593:9372
- fd22:4e39:e630:1:7039:14cd:ab23:6cb0
- fd22:4e39:e630:1:ac06:2b2d:9818:4925

Public Address:

- fd22:4e39:e630:1:ae7f:28f9:781f:111a

Link-Local Address:

- fe80::5d20:af94:8779:1efd%5

Analysis

IPv6 assigned multiple addresses to the Wi-Fi interface. Temporary addresses improve privacy by preventing long-term tracking, while the link-local address allows communication within the local network segment.

Wireshark Portion

Results

Wireshark was not installed, and packet captures were not collected.

Analysis

The IPv6 packet-capture and filtering procedures described in the lab could not be performed. Consequently, IPv6 traffic could not be verified using Wireshark.

Problems Encountered

1. The command ipconfig /displaydns required administrative privileges and therefore could not be completed.
2. Lab B Step 3 was not completed.
3. Wireshark was not installed, so IPv6 packet capture and analysis were not performed.

Conclusion

This lab provided practical experience using networking utilities including ipconfig, ping, nslookup, netstat, and netsh. Hostname resolution was examined through both localhost and custom hosts file entries. IPv4 and IPv6 loopback functionality were successfully tested, demonstrating proper operation of both protocol stacks. The lab also introduced IPv6 addressing concepts and showed how operating systems use local configuration files and DNS services to resolve hostnames. Although some portions of the lab were not completed due to administrative permission requirements and the absence of Wireshark, the objectives related to network configuration and connectivity testing were successfully achieved.

```
C:\Users\dstdi>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : LAPTOP-Q8KU3PPL
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-07
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::e83a:15c5:ed3:dcd2%7(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 671744039
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2E-94-E2-4B-F8-54-F6-23-C1-78
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : FA-54-F6-23-C1-78
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : FE-54-F6-23-C1-78
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : home
   Description . . . . . . . . . . . : Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
```

```
C:\Users\dstdi>ping ::1

Pinging ::1 with 32 bytes of data:
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms
Reply from ::1: time<1ms

Ping statistics for ::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\Users\dstdi>nslookup
Default Server:  openrg.home
Address:  fd22:4e39:e630:1:2ede:32ff:fee2:86ff

> help
Commands:   (identifiers are shown in uppercase, [] means optional)
NAME            - print info about the host/domain NAME using default server
NAME1 NAME2     - as above, but use NAME2 as server
help or ?       - print info on common commands
set OPTION      - set an option
    all                 - print options, current server and host
    [no]debug           - print debugging information
    [no]d2              - print exhaustive debugging information
    [no]defname         - append domain name to each query
    [no]recurse         - ask for recursive answer to query
    [no]search          - use domain search list
    [no]vc              - always use a virtual circuit
    domain=NAME         - set default domain name to NAME
    srchlist=N1[/N2/.../N6] - set domain to N1 and search list to N1,N2, etc.
    root=NAME           - set root server to NAME
    retry=X             - set number of retries to X
    timeout=X           - set initial time-out interval to X seconds
    type=X              - set query type (ex. A,AAAA,A+AAAA,ANY,CNAME,MX,NS,PTR,SOA,SRV)
    querytype=X         - same as type
    class=X             - set query class (ex. IN (Internet), ANY)
    [no]msxfr           - use MS fast zone transfer
    ixfrver=X           - current version to use in IXFR transfer request
server NAME     - set default server to NAME, using current default server
lserver NAME    - set default server to NAME, using initial server
root            - set current default server to the root
ls [opt] DOMAIN [> FILE] - list addresses in DOMAIN (optional: output to FILE)
    -a          -  list canonical names and aliases
    -d          -  list all records
    -t TYPE     -  list records of the given RFC record type (ex. A,CNAME,MX,NS,PTR etc.)
view FILE           - sort an 'ls' output file and view it with pg
exit            - exit the program
```

```
Displays protocol statistics and current TCP/IP network connections.

NETSTAT [-a] [-b] [-e] [-f] [-i] [-n] [-o] [-p proto] [-r] [-s] [-t] [-x] [-y] [interval]

  -a            Displays all connections and listening ports.
  -b            Displays the executable involved in creating each connection or
                listening port. In some cases well-known executables host
                multiple independent components, and in these cases the
                sequence of components involved in creating the connection
                or listening port is displayed. In this case the executable
                name is in [] at the bottom, on top is the component it called,
                and so forth until TCP/IP was reached. Note that this option
                can be time-consuming and will fail unless you have sufficient
                permissions.
  -c            Displays a list of processes sorted by the number of TCP or UDP
                ports currently consumed.
  -d            Displays DSCP value assocated with each connection.
  -e            Displays Ethernet statistics. This may be combined with the -s
                option.
  -f            Displays Fully Qualified Domain Names (FQDN) for foreign
                addresses.
  -i            Displays the time spent by a TCP connection in its current state.
  -n            Displays addresses and port numbers in numerical form.
  -o            Displays the owning process ID associated with each connection.
  -p proto      Shows connections for the protocol specified by proto; proto
                may be any of: TCP, UDP, TCPv6, or UDPv6.  If used with the -s
                option to display per-protocol statistics, proto may be any of:
                IP, IPv6, ICMP, ICMPv6, TCP, TCPv6, UDP, or UDPv6.
  -q            Displays all connections, listening ports, and bound
                nonlistening TCP ports. Bound nonlistening ports may or may not
                be associated with an active connection.
  -r            Displays the routing table.
  -s            Displays per-protocol statistics.  By default, statistics are
                shown for IP, IPv6, ICMP, ICMPv6, TCP, TCPv6, UDP, and UDPv6;
                the -p option may be used to specify a subset of the default.
  -t            Displays the current connection offload state.
  -x            Displays NetworkDirect connections, listeners, and shared
                endpoints.
  -y            Displays the TCP connection template for all connections.
                Cannot be combined with the other options.
  interval      Redisplays selected statistics, pausing interval seconds
                between each display.  Press CTRL+C to stop redisplaying
                statistics.  If omitted, netstat will print the current
                configuration information once.
```

```
C:\Users\dstdi>netsh interface ipv6 show address

Interface 1: Loopback Pseudo-Interface 1

Addr Type  DAD State   Valid Life Pref. Life Address
---------  ----------- ---------- ---------- ------------------------
Other      Preferred     infinite   infinite ::1

Interface 5: Wi-Fi

Addr Type  DAD State   Valid Life Pref. Life Address
---------  ----------- ---------- ---------- ------------------------
Temporary  Preferred   5d14h22m4s 5d14h19m28s fd22:4e39:e630:1:965:548f:a593:9372
Temporary  Preferred  5d19h20m53s 5d19h18m17s fd22:4e39:e630:1:7039:14cd:ab23:6cb0
Temporary  Preferred  5d13h13m27s 5d13h10m51s fd22:4e39:e630:1:ac06:2b2d:9818:4925
Public     Preferred     infinite   infinite fd22:4e39:e630:1:ae7f:28f9:781f:111a
Other      Preferred     infinite   infinite fe80::5d20:af94:8779:1efd%5

Interface 14: Local Area Connection* 1

Addr Type  DAD State   Valid Life Pref. Life Address
---------  ----------- ---------- ---------- ------------------------
Other      Deprecated    infinite   infinite fe80::767d:a20c:716:f0fb%14

Interface 17: Bluetooth Network Connection

Addr Type  DAD State   Valid Life Pref. Life Address
---------  ----------- ---------- ---------- ------------------------
Other      Deprecated    infinite   infinite fe80::b000:c32e:8c45:cc26%17

Interface 15: Local Area Connection* 2

Addr Type  DAD State   Valid Life Pref. Life Address
---------  ----------- ---------- ---------- ------------------------
Other      Deprecated    infinite   infinite fe80::5c77:2205:3f6d:fdd%15

Interface 7: Ethernet

Addr Type  DAD State   Valid Life Pref. Life Address
---------  ----------- ---------- ---------- ------------------------
Other      Preferred     infinite   infinite fe80::e83a:15c5:ed3:dcd2%7
```